Deep Learning (23AID211)

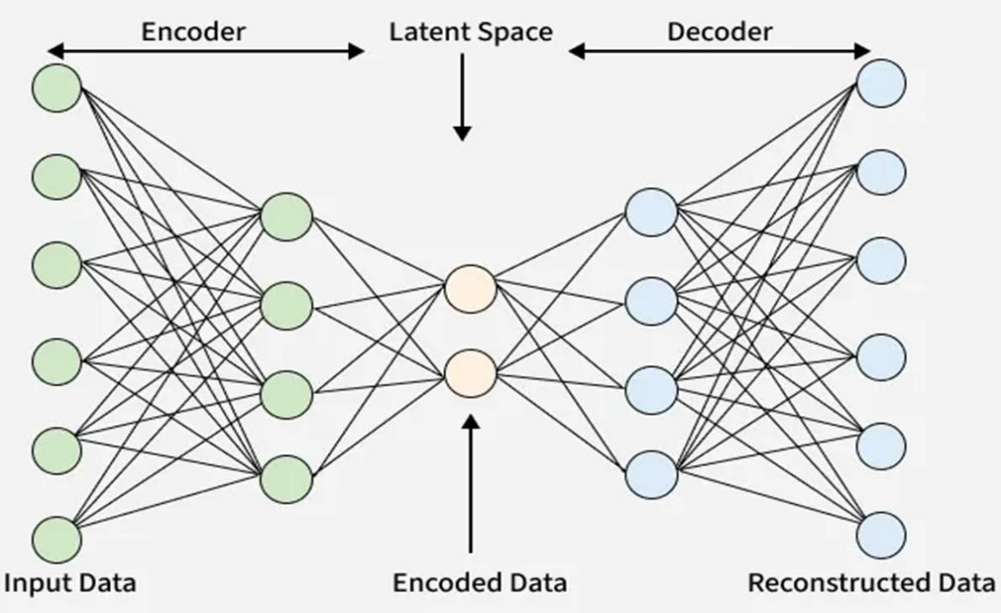
Autoencoders (AEs)

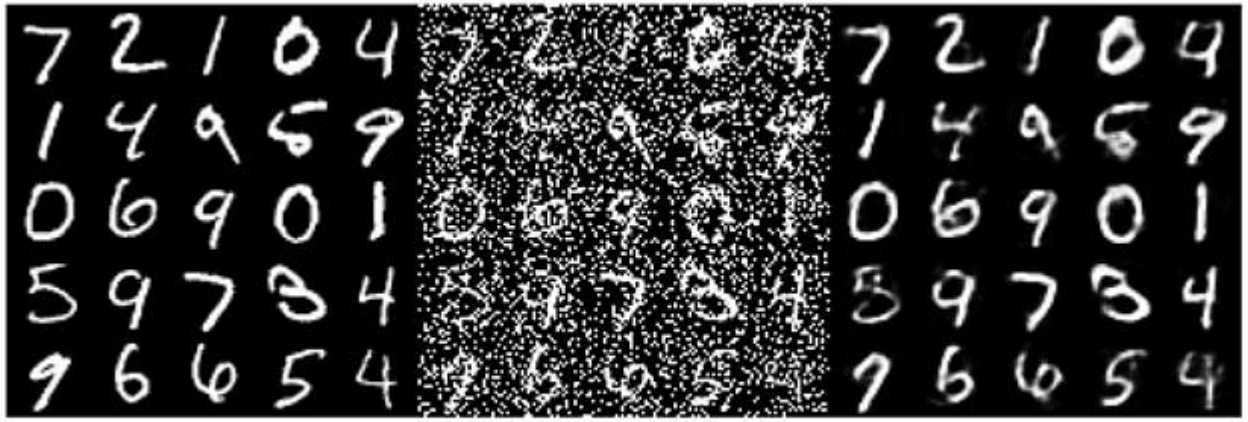
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Auto Encoders (AEs)

- An autoencoder is an unsupervised neural network that learns to compress data into a lower-dimensional latent representation (encoding) and then reconstructs the original input (decoding) with minimum error.

- It consists of an encoder, a bottleneck (latent space), and a decoder, making it valuable for dimensionality reduction, feature extraction, denoising, and anomaly detection.



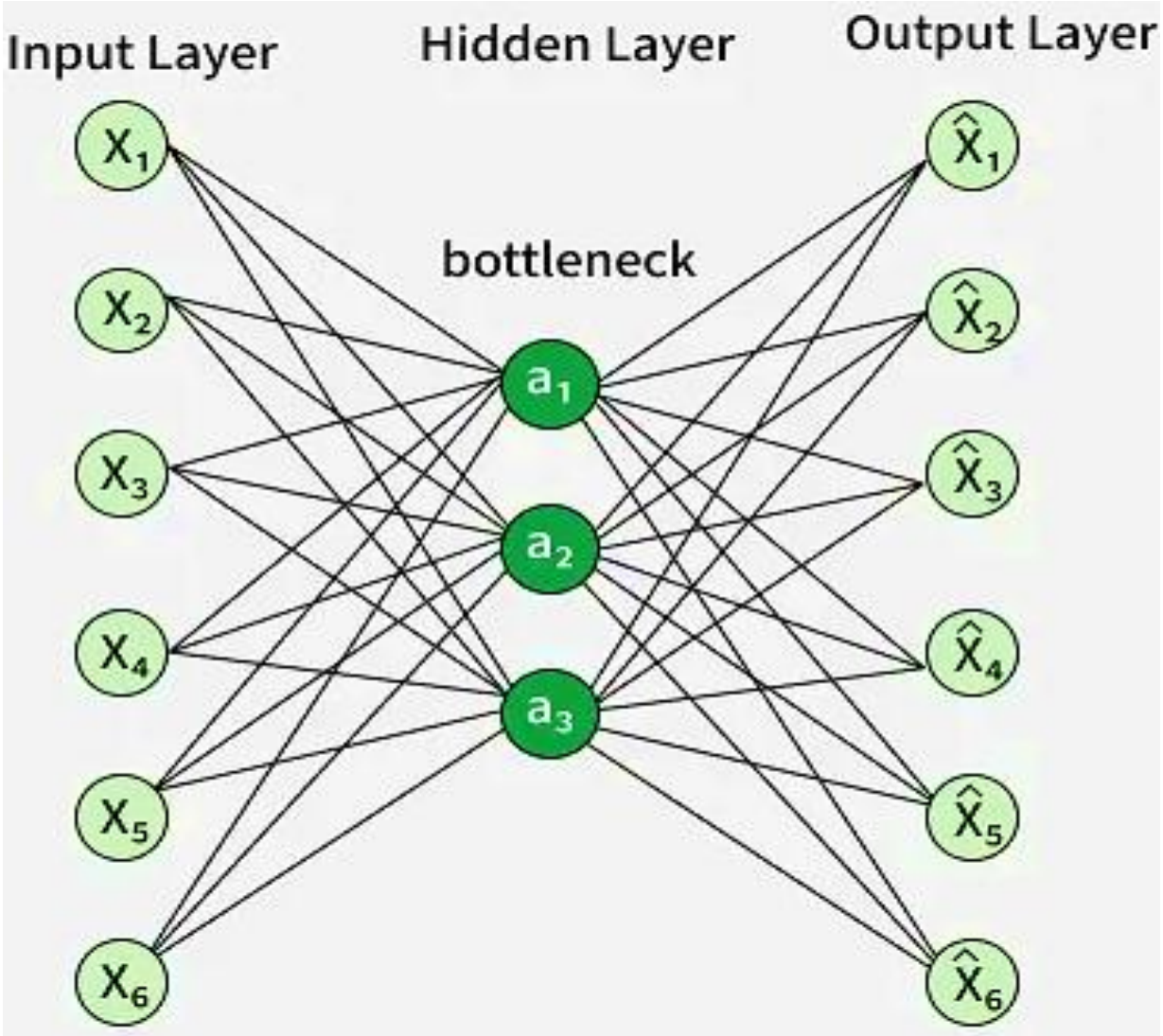


Original input, corrupted data, reconstructed data

Auto Encoders

- An autoencoder is an unsupervised neural network that compresses data into lower-dimensional latent representation as encoding and then reconstructs the original input as decoding with minimum error.
- An autoencoder's architecture consists of three main components as follows:

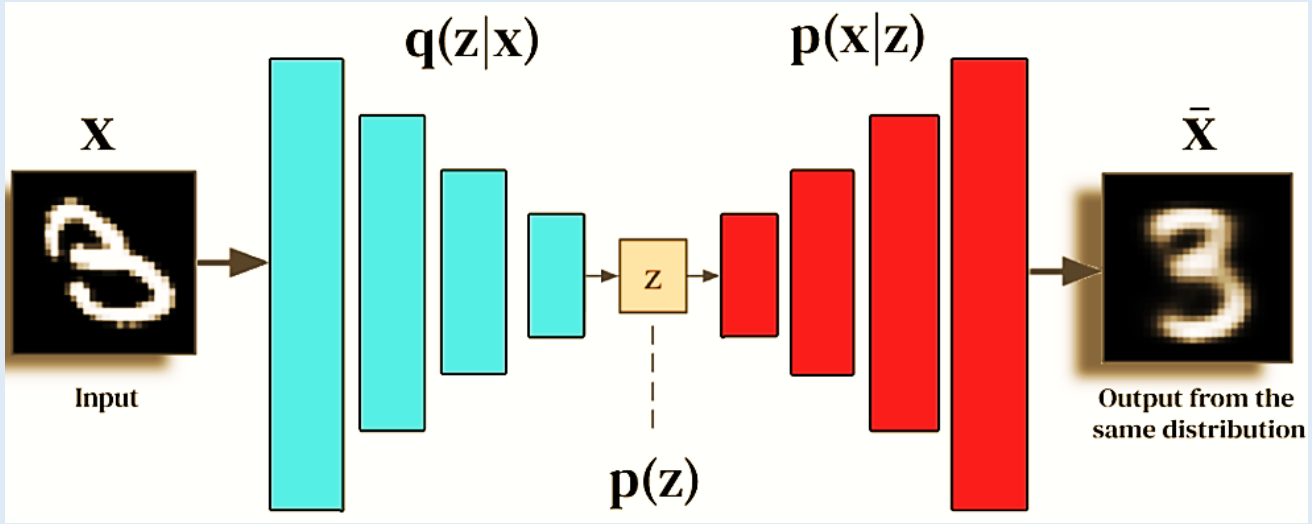
1. Encoder
2. Bottleneck (Latent Space)
3. Decoder



Types of Autoencoders

Over the years, different types of Autoencoders have been developed:

- Vanilla Autoencoder
- Sparse Autoencoder
- Undercomplete Autoencoder
- Denoising Autoencoder
- Contractive Autoencoder
- Convolutional Autoencoder
- Variational Autoencoder



1. Vanilla Autoencoder

- Vanilla Autoencoder are the simplest form used for unsupervised learning tasks. They consist of two main parts an encoder that compresses the input data into a smaller, dense representation and a decoder that reconstructs the original input from this compressed form.
- Training minimizes reconstruction error which measures the difference between input and output. This optimization is done via backpropagation which helps in updating the network weights to improve reconstruction accuracy.
- They are foundational models helps in serving as building blocks for more complex variants.

2. Sparse Autoencoder

- A Sparse Autoencoder is quite similar to an Undercomplete Autoencoder, but their main difference lies in how regularization is applied.
- With Sparse Autoencoders, we don’t necessarily need to reduce the bottleneck dimensionality.
- This penalty is commonly referred to as a **sparsity function**, and it's quite different from traditional regularization techniques.

Methods to Enforce Sparsity

Sparsity is enforced to ensure the autoencoder learns high-level features by activating only a small subset of neurons

- **L1 Regularization:** Applies a penalty proportional to the absolute activations or weights, encouraging the network to use fewer neurons.
- **KL Divergence:** Measures how far the average activation deviates from the target sparsity level, ensuring that only a subset of neurons activate at a time.
- **Sparsity Proportion:** Defines the desired activation frequency of hidden units across training samples.

Training of Sparse Autoencoder

1. **Initialization:** The network weights are initialized randomly or using pre-trained models to provide a stable starting point.
2. **Forward Pass:** Input data is passed through the encoder to obtain the latent (compressed) representation and then through the decoder to reconstruct the original input.
3. **Loss Calculation:** The total loss combines:
   - **Reconstruction error:** MSE between X and $\hat{X}$
   - **Sparsity penalty:** Uses KL Divergence or L1 regularization to keep most hidden units inactive.

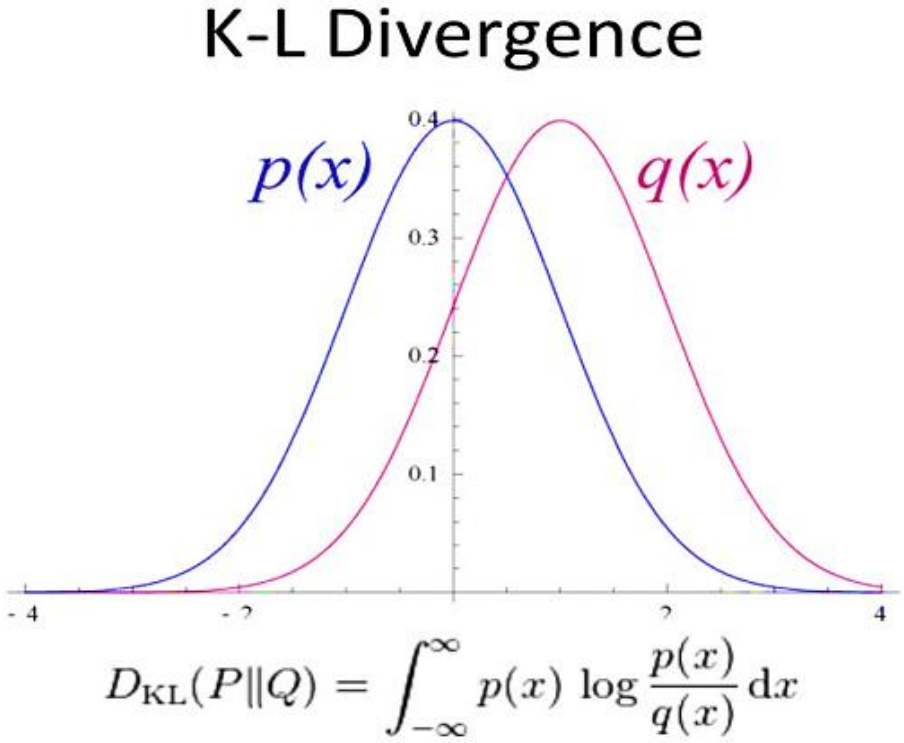
Kullback Leibler (KL Divergence)

Sparse autoencoders combine an encoder, a decoder, and a sparsity-constrained loss function to learn compact, meaningful representations. The loss function used is:

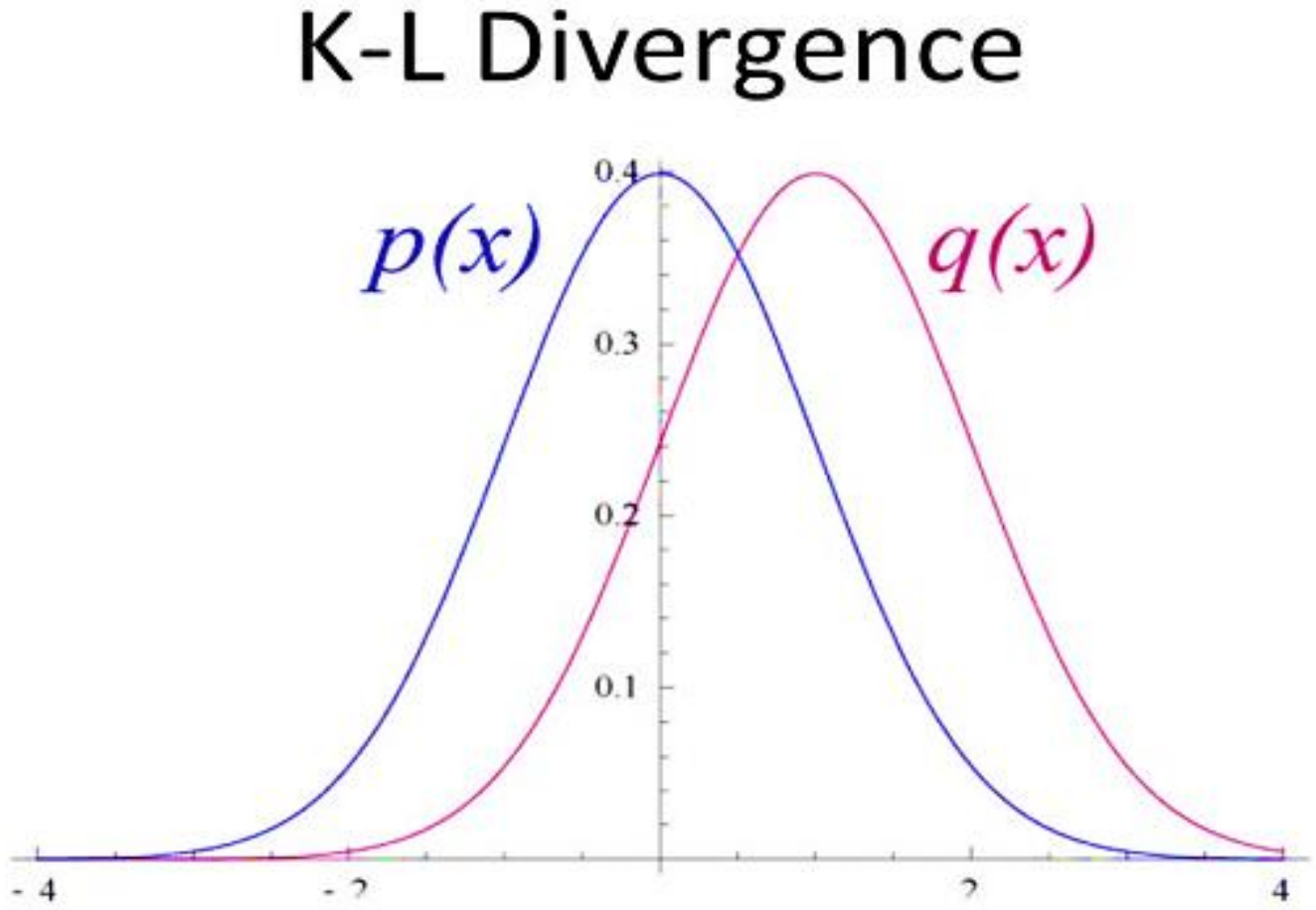
$$L = \|X - \hat{X}\|^2 + \lambda \cdot \text{Penalty}(s)$$

where

- X: Input data
- $\hat{X}$: Reconstructed output
- λ: Regularization parameter
- $\text{Penalty}(s)$: A function penalizing deviations from sparsity



Kullback Leibler (KL Divergence)



$$D_{\mathrm{KL}}(P\|Q) = \int_{-\infty}^{\infty} p(x)\,\log\frac{p(x)}{q(x)}\,\mathrm{d}x$$

3. Denoising Autoencoder

- Denoising Autoencoders are designed to handle corrupted or noisy inputs by learning to reconstruct the clean, original data.
- Training involves feeding intentionally corrupted inputs and minimizing the reconstruction error against the clean version.
- This approach forces the model to capture robust features that are invariant to noise.

- A Denoising Autoencoder (DAE) is a type of neural network that learns to remove noise from corrupted input data by reconstructing the original, clean data.
- It enhances feature learning by adding noise (e.g., Gaussian or Salt-and-Pepper) to the input, forcing the model to learn robust representations rather than simply copying the input.

4. Undercomplete Autoencoder

- This is the simplest version of an autoencoder.
- In this case, we don’t have an explicit regularization mechanism, but we ensure that the bottleneck size is always smaller than the original input size to avoid overfitting.
- This type of configuration is typically used as a dimensionality reduction technique (more powerful than PCA, since it can also capture non-linearities in the data).
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4. Architectural Differences Between Undercomplete and Undercomplete Autoencoder

- **Vanilla Autoencoder:** Typically refers to a simple 3-layer neural network (input, hidden, output) where the hidden layer (code) dimension can be anything.

- **Undercomplete Autoencoder:** Specifically requires the **dimension of the hidden layer to be smaller than the dimension of the input layer**. This creates a "bottleneck" that forces compression.

5. Contractive Autoencoder

- Contractive Autoencoders introduce an additional penalty during training to make the learned representations robust to small changes in input data.
- They minimize both reconstruction error and a regularization term that penalizes sensitivity to input perturbations.
- This results in stable, invariant features useful in noisy or fluctuating environments.

6. Convolutional Autoencoder

- Convolutional Autoencoders use convolutional layers to effectively capture spatial and hierarchical features in high-dimensional data such as images.
- These models optimize reconstruction error using loss functions suited for images like mean squared error or binary cross-entropy.
- The architecture helps in handling structured inputs by preserving spatial relationships.

7. Variational Autoencoder

- Variational Autoencoder (VAEs) extend traditional autoencoders by learning probabilistic latent distributions instead of fixed representations. Training optimizes the Evidence Lower Bound (ELBO), which balances:
- Reconstruction loss to ensure accurate data reconstruction.
- KL Divergence to regularize the latent space towards a standard Gaussian helps in preventing overfitting and smooth latent structure.
- By balancing these two terms VAEs can generate meaningful outputs while keeping the latent space structured.